Notes: Kandel and Pearson (JPE, 1995)

Differential Interpretation of Public Signals and Trade in Speculative Markets

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1 One-sentence summary

This note summarizes Kandel and Pearson (JPE, 1995), which uses CRSP and Compustat earnings-announcement windows, abnormal trading-volume tests by announcement-return cells, a speculative-trade model with heterogeneous likelihood functions, and I/B/E/S analyst forecast revisions to show that public signals generate trade even when prices barely move, and that analysts' belief revisions display flips and divergences that are difficult to reconcile with identical interpretations of public information.

2 Big picture

2.1 Core question

Question. When all traders observe the same public announcement, do they interpret the information in the same way?

The paper argues that the standard assumption of identical interpretation is too restrictive. In many speculative-market models, agents may have different private signals, but they share the same likelihood function: they agree on how signals map into fundamentals. Kandel and Pearson challenge this by asking whether investors and analysts may instead use different models of the world to interpret the same public signal.

2.2 Why this matters

If agents interpret public information identically, then trade around public announcements is tightly linked to price changes. In the canonical logic, a public announcement that does not move the price should not create much information-based trade. If trade is high even when prices do not move, then the announcement may be generating disagreement rather than a common revision of beliefs.

The paper therefore connects three themes:

- **Information aggregation:** prices may reveal average beliefs but hide disagreement.
- **Speculative trade:** trade can arise from disagreement about how to interpret a common signal.
- **Belief heterogeneity:** differences in likelihood functions can be persistent, not merely temporary private information.

2.3 Main finding

The paper documents two related facts. First, earnings announcements generate significant abnormal trading volume in every return cell, including the zero-return cell. Second, analyst forecasts around earnings announcements contain flips and divergences - revisions that should not occur under identical likelihoods but can occur when analysts interpret the same announcement differently.

3 Incremental contribution of the paper

3.1 Contribution relative to standard trading-volume models

Earlier trading models explain speculative trade using private information or heterogeneous prior beliefs, but usually assume that agents interpret public signals using a common likelihood function. Kandel and Pearson's first contribution is to show that this structure implies a sharp volume-return prediction: announcement-induced trade should be closely tied to the magnitude of the announcement return.

Empirically, however, earnings announcements produce abnormal volume even when the announcement return is approximately zero. This shifts the focus from **"what private information did each agent have?"** to **"how did different agents interpret the same public information?"**

3.2 Contribution relative to disagreement models

The paper does not simply say that agents disagree. It models a precise form of disagreement: agents use different likelihood functions to interpret the public signal. This generates a volume equation with an intercept term:

$$V = |\beta_0 + \beta_1 \Delta P^*|,$$

where β_0 is nonzero when agents interpret the public signal differently. The key implication is that volume can be positive even when the price change ΔP^* is zero.

3.3 Contribution through belief data

The paper also uses analyst forecast revisions as direct evidence on beliefs. Instead of inferring disagreement only from prices and volume, it studies how named analysts revise earnings forecasts around the same public announcements. The forecast-revision tests give the paper a second layer of evidence: not only do markets trade as if interpretations differ, but analysts' reported beliefs also move in ways that suggest differential interpretation.

4 Conceptual framework

4.1 Standard view: identical interpretation

Most speculative-trade models allow investors to begin with different information, but impose a shared mapping from signals to fundamentals. Under this view, a public signal should make agents' beliefs more aligned. If an announcement has no price impact, it should not cause large disagreement-driven trading.

The paper highlights the benchmark implication from Kim and Verrecchia-type models:

$$V_{it} = k_{it}|\Delta P_{it}|,$$

where V_{it} is information-related trading volume around announcement t for security i, ΔP_{it} is the price change, and k_{it} summarizes heterogeneity in prior precision and risk tolerance. The crucial implication is:

$$\Delta P_{it} = 0 \quad \Rightarrow \quad V_{it} = 0.$$

4.2 Alternative view: differential interpretation

Kandel and Pearson propose that agents may observe the same signal but use different likelihood functions. The public signal is

$$L = X + \varepsilon,$$

where X is the asset payoff and ε is noise. Agents agree that the signal is public but disagree about the distribution of ε. If type i believes $\varepsilon \sim N(\mu_i, b_i^{-1})$, then different μ_i's mean that the same signal is read optimistically by one agent and pessimistically by another.

With normal beliefs and exponential utility, the posterior mean of type i is

$$Y_i = \rho_i X_i + (1 - \rho_i)(L - \mu_i),$$

where X_i is the prior mean and ρ_i is the weight placed on the prior. If two agents have different μ_i's, the same public announcement can move their posterior beliefs in different directions.

4.3 Economic intuition

A public announcement can have two effects:

- **Common-value effect:** it shifts the average market belief and therefore the price.
- **Disagreement effect:** it changes the dispersion of beliefs across agents.

Identical interpretation allows the first effect but severely restricts the second. Differential interpretation allows both. The announcement may leave the average belief nearly unchanged while still causing large disagreement, portfolio reallocation, and trading volume.

5 Empirical design I: volume-return relation

5.1 Data

The first empirical exercise studies quarterly earnings announcements. The announcement dates come from Compustat quarterly files and cover June 1981 through June 1992. Returns and trading volumes come from CRSP for NYSE/ASE stocks. The final NYSE/ASE sample contains 2,483 firms and 69,067 firm-announcements.

5.2 Announcement and nonannouncement windows

The paper uses a three-day announcement window, days $-1, 0, +1$, because Compustat's announcement date may not always be the exact day on which investors first observe the information.

Volume is standardized by average trading volume during days -46 through -17. The normal comparison volume is built from preannouncement three-day periods and is matched by return cells, so the abnormal volume comparison does not simply reflect the known positive relation between absolute returns and volume.

5.3 Return-cell design

The authors partition events into 23 return cells. The middle cell is the approximately zero-return cell. This design asks:

> For the same realized announcement return, is volume higher during the announcement window than during nonannouncement windows?

The most important cell is the zero-return cell. Identical-interpretation models predict little or no information-related volume there, while differential-interpretation models predict positive volume if agents disagree about what the signal means.

6 Main volume evidence

6.1 Table 1 and Figure 1: announcement volume is high in every return cell

- **Read:** Table 1 reports mean and median volumes for announcement and nonannouncement windows by return cell. Figure 1 plots the median volume-return relation.
- **Key pattern:** announcement volume is above nonannouncement volume in every return cell.
- **Most important point:** even in the zero-return cell, announcement volume is abnormally high.
- **Economic message:** volume is not merely a mechanical function of price movement. Public news can create trade even when the average price response is small.

TABLE 1

MEAN AND MEDIAN ANNOUNCEMENT AND NONANNOUNCEMENT VOLUMES FOR NYSE/ASE STOCKS

		Nonannouncement Sample				Announcement Sample				
Cell	Return Range	Observations	Mean Volume	Standard Error	Median Volume	Observations	Mean Volume	Standard Error	Median Volume	Wilcoxon p-Value
1	$R < -.05$	29,319	1.562	.022	.991	8,849	2.213	.039	1.430	<.0001
2	$-.045 > R \geq -.05$	5,916	1.186	.030	.839	1,286	1.526	.086	1.050	<.0001
3	$-.040 > R \geq -.045$	6,325	1.127	.021	.807	1,478	1.456	.056	1.058	<.0001
4	$-.035 > R \geq -.040$	8,481	1.198	.115	.801	1,734	1.655	.103	1.062	<.0001
5	$-.030 > R \geq -.035$	10,127	1.059	.014	.783	1,952	1.392	.040	.975	<.0001
6	$-.025 > R \geq -.030$	11,586	1.036	.012	.765	2,196	1.370	.061	.959	<.0001
7	$-.020 > R \geq -.025$	14,013	1.005	.015	.732	2,455	1.332	.036	.942	<.0001
8	$-.015 > R \geq -.020$	16,943	1.031	.018	.739	2,929	1.235	.023	.933	<.0001
9	$-.010 > R \geq -.015$	19,062	.981	.012	.719	3,111	1.251	.026	.904	<.0001
10	$-.005 > R \geq -.010$	20,386	.981	.015	.720	3,404	1.207	.027	.892	<.0001
11	$-.00001 \geq R \geq -.005$	9,034	.968	.016	.733	1,414	1.219	.045	.915	<.0001
12	$.00001 > R > -.00001$	45,489	.942	.008	.655	7,209	1.302	.035	.831	<.0001
13	$.005 \geq R \geq .00001$	8,848	1.016	.021	.746	1,440	1.326	.071	.922	<.0001
14	$.010 \geq R > .005$	19,757	1.008	.014	.749	3,348	1.456	.128	.921	<.0001
15	$.015 \geq R > .010$	17,827	1.030	.010	.784	3,061	1.305	.026	.949	<.0001
16	$.020 \geq R > .015$	15,870	1.103	.014	.820	2,859	1.409	.039	1.015	<.0001
17	$.025 \geq R > .020$	12,871	1.126	.016	.852	2,316	1.432	.032	1.065	<.0001
18	$.030 \geq R > .025$	10,703	1.228	.024	.904	2,116	1.619	.046	1.136	<.0001
19	$.035 \geq R > .030$	9,110	1.257	.035	.905	1,895	1.568	.038	1.188	<.0001
20	$.040 \geq R > .035$	7,783	1.303	.019	.969	1,664	2.063	.476	1.163	<.0001
21	$.045 \geq R > .040$	6,253	1.352	.020	1.017	1,371	1.704	.045	1.270	<.0001
22	$.050 \geq R > .045$	5,708	1.539	.134	1.015	1,337	1.742	.046	1.272	<.0001
23	$R > .05$	33,981	2.011	.021	1.315	9,643	3.042	.094	1.797	<.0001
Total		345,392				69,067				

NOTE.—Events for which volume data are available for fewer than seven of the 10 3-day periods before day −16 are excluded.

Table 1: Announcement and nonannouncement volumes by return cell

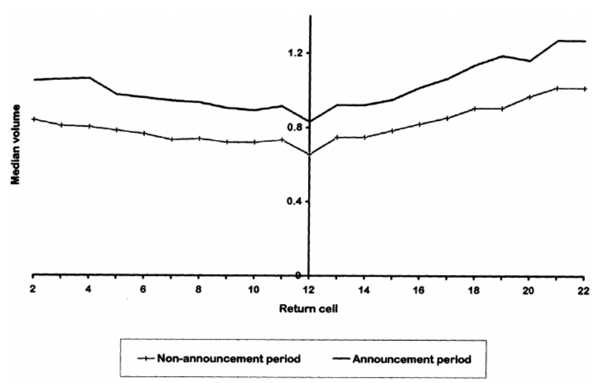

FIG. 1.—Median announcement and nonannouncement period volumes for NYSE/ASE stocks, by return cell. Events for which volume data are available for fewer than seven of the 10 3-day periods before day −16 are excluded.

Figure 1: Median volume-return relation around announcements

6.2 Interpretation of the zero-return evidence

The zero-return evidence is the cleanest challenge to standard models. If all agents interpret the announcement in the same way, and the announcement produces essentially no price change, then there should be little reason for large information-motivated reallocation. Yet Table 1 and Figure 1 show that trading volume remains high. This is consistent with an announcement that creates offsetting revisions: some agents interpret the signal positively and buy, while others interpret it negatively and sell, leaving the price almost unchanged but generating trade.

6.3 Table 2 and Figure 2: liquidity-shifting explanation is weak

One alternative explanation is that liquidity traders postpone trading until after the announcement because preannouncement information asymmetry is high. If so, volume should be unusually low just before the announcement. Table 2 and Figure 2 examine two preannouncement windows, days $-4, -2$ and days $-7, -5$. They show little evidence of depressed preannouncement volume. This weakens the liquidity-shifting explanation.

TABLE 2

MEAN VOLUMES FOR NYSE/ASE STOCKS FOR THE NONANNOUNCEMENT PERIOD AND TWO 3-DAY PERIODS JUST BEFORE THE ANNOUNCEMENT

		NONANNOUNCEMENT SAMPLE		DAYS −4 THROUGH −2			DAYS −7 THROUGH −5		
CELL	RETURN RANGE	Observations	Mean Volume	Observations	Mean Volume	Wilcoxon p-Value	Observations	Mean Volume	Wilcoxon p-Value
1	$R < -.05$	29,319	1.562	6,142	1.655	.0001	5,963	1.550	.0003
2	$-.045 > R \geq -.05$	5,916	1.186	1,183	1.193	.2050	1,210	1.129	.9526
3	$-.040 > R \geq -.045$	6,325	1.127	1,271	1.188	.1150	1,287	1.115	.4037
4	$-.035 > R \geq -.040$	8,481	1.198	1,703	1.156	.2632	1,670	1.085	.4292
5	$-.030 > R \geq -.035$	10,127	1.059	1,957	1.092	.1032	2,039	1.044	.8400
6	$-.025 > R \geq -.030$	11,586	1.036	2,308	1.099	.0411	2,272	1.067	.0570
7	$-.020 > R \geq -.025$	14,013	1.005	2,814	.996	.1118	2,869	1.033	.5813
8	$-.015 > R \geq -.020$	16,943	1.031	3,386	1.025	.4050	3,317	1.054	.0205
9	$-.010 > R \geq -.015$	19,062	.981	3,732	.982	.3632	3,789	.999	.9544
10	$-.005 > R \geq -.010$	20,386	.981	3,941	.974	.0449	4,025	.990	.0844
11	$-.00001 \geq R \geq -.005$	9,034	.968	1,759	.958	.4180	1,780	.980	.5228
12	$.00001 > R > -.00001$	45,489	.942	9,105	.979	.0089	9,125	.983	.1149
13	$.005 \geq R \geq .00001$	8,848	1.016	1,724	.996	.6901	1,827	1.058	.4455
14	$.010 \geq R > .005$	19,757	1.008	3,890	1.062	.0841	3,938	.984	.8671
15	$.015 \geq R > .010$	17,827	1.030	3,456	1.067	.3459	3,536	1.027	.8108
16	$.020 \geq R > .015$	15,870	1.103	3,164	1.103	.9845	3,138	1.145	.7133
17	$.025 \geq R > .020$	12,871	1.126	2,580	1.153	.0142	2,585	1.180	.1781
18	$.030 \geq R > .025$	10,703	1.228	2,105	1.264	.2820	2,156	1.229	.3054
19	$.035 \geq R > .030$	9,110	1.257	1,854	1.305	.3210	1,837	1.348	.9103
20	$.040 \geq R > .035$	7,783	1.303	1,646	1.358	.1889	1,597	1.332	.6449
21	$.045 \geq R > .040$	6,253	1.352	1,224	1.404	.2255	1,266	1.477	.0016
22	$.050 \geq R > .045$	5,708	1.539	1,093	1.394	.7139	1,159	1.542	.3377
23	$R > .05$	33,981	2.011	7,048	2.177	.0003	6,699	2.110	.0005
Total		345,392		69,085			69,084		

NOTE.—Events for which volume data are available for fewer than seven of the 10 3-day periods before day −16 are excluded.

Table 2: Volumes before the announcement window

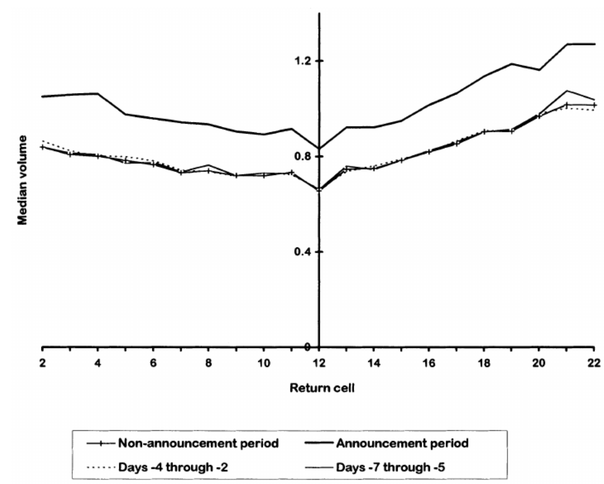

Figure 2: Median volumes in preannouncement and announcement windows

7 Alternative explanations considered by the paper

The authors discuss several ways one might try to explain the abnormal volume without differential interpretation.

7.1 Kim and Verrecchia-style proportionality

Under the proportional volume-price relation, information-related volume should vanish when the price change is zero. This does not match the zero-return-cell evidence.

7.2 Harris and Raviv-style disagreement

Harris and Raviv also study opinion differences, but their model implies that trade around public signals is accompanied by price changes. Therefore, the model does not naturally explain abnormal volume when returns are close to zero.

7.3 Liquidity trading shifted to announcement days

If liquidity traders delay trade until after news is released, announcement-day volume can be high. But the paper shows that volume just before announcements is not unusually low. This explanation therefore does not fit the timing pattern.

7.4 Mismeasured announcement returns

Perhaps a three-day return of zero hides large intraday movements that net out. The authors repeat the analysis using one-day windows and obtain similar results, suggesting that the evidence is not driven by poor return measurement.

7.5 Private information production

Another possibility is that private information production is concentrated around earnings announcements. The authors cannot rule this out completely. But they argue that private information production would not naturally be restricted to the narrow three-day announcement window, except under mechanisms close to differential interpretation itself.

7.6 Fully revealing equilibrium, wealth changes, risk shifts, and clientele shifts

The paper also considers whether announcements trigger a move from partially revealing to fully revealing equilibria, wealth-based trading, risk reallocation, or clientele shifts. The authors argue that none of these explanations comfortably matches the combination of abnormal volume, zero-return-cell evidence, and the analyst forecast results.

8 The different likelihoods model

8.1 Setup

The model has a risky asset and a riskless asset. There are two types of agents. Each type begins with a prior belief about the payoff X, observes the same public signal L, and updates beliefs using its own likelihood function. The signal is

$$L = X + \varepsilon,$$

but agents disagree about the distribution of ε. In particular, each type believes the signal error has a different mean μ_i and precision b_i.

8.2 Trading volume implication

After the announcement, the change in holdings is linearly related to the price change, but with an intercept:

$$V = |\beta_0 + \beta_1 \Delta P^*|.$$

The intercept β_0 is proportional to differences in likelihood means, especially $\mu_2 - \mu_1$. Therefore:

$$\mu_1 \neq \mu_2 \quad \Rightarrow \quad \beta_0 \neq 0.$$

This is the key theoretical result. If agents interpret the same public signal differently, trade can occur even when the price change is zero. If agents have identical likelihood functions, $\beta_0 = 0$, and the model collapses toward the standard proportional volume-price prediction.

9 Empirical design II: analyst forecast revisions

9.1 Why use analyst forecasts?

Volume and prices reveal behavior, but analyst forecasts provide a direct measure of beliefs. The paper uses I/B/E/S analyst earnings forecasts to observe how individual analysts revise beliefs around the same public earnings announcement.

The sample uses firms followed by at least 10 analysts at some point during 1983-1991. The authors focus on forecasts of current fiscal-year earnings around first-, second-, and third-quarter earnings announcements.

9.2 The theoretical restriction under identical likelihoods

Let X_i and X_j be two analysts' prior forecasts, and Y_i and Y_j their posterior forecasts after the public announcement. Under identical likelihoods, the posterior forecasts cannot behave in certain ways when analysts revise in opposite directions.

The paper defines two inconsistent revision patterns.

Flip:

$$\operatorname{sgn}(Y_i - X_i) \neq \operatorname{sgn}(Y_j - X_j), \quad X_i > X_j, \quad Y_i < Y_j.$$

A flip occurs when the analyst who was initially more optimistic becomes more pessimistic after the announcement.

Divergence:

$$\operatorname{sgn}(Y_i - X_i) \neq \operatorname{sgn}(Y_j - X_j), \quad |Y_i - Y_j| > |X_i - X_j|.$$

A divergence occurs when analysts revise in opposite directions and end farther apart than before.

Under identical likelihoods, these patterns should not occur. Under differential interpretation, they can occur.

9.3 Table 3: announcement and control windows

Table 3 defines the windows used to compare announcement-period revisions with nonannouncement-period revisions. The point is to isolate belief changes plausibly caused by public earnings news and to compare them to revisions in control windows.

TABLE 3

WINDOWS USED IN COMPARISONS OF THE FREQUENCIES OF FLIPS AND DIVERGENCES

	ANNOUNCEMENT SAMPLES		NONANNOUNCEMENT SAMPLES	
COMPARISON	Window 1 (A1)	Window 2 (A2)	Window 1 (NA1)	Window 2 (NA2)
1	days −13–0	days 1–28	days 29–42	days 43–70
2	days −27–0	days 1–56	days 1–28	days 29–84
3	days −41–0	days 1–42	days 1–42	days 43–84

NOTE.—Days are measured in calendar days relative to the interim earnings announcement date, which is day 0.

Table 3: Forecast-revision windows

TABLE 4

AVERAGE PROPORTIONS OF EXPLICIT FORECAST REVISIONS

	ALL ANALYST PAIRS			ANALYST PAIRS WITH SMALL CHANGES IN AVERAGE FORECAST			ANALYST PAIRS WITH FORECAST REVISIONS OF OPPOSITE SIGN		
SAMPLE	Flips (1)	Divergences (2)	Inconsistencies (3)	Flips (4)	Divergences (5)	Inconsistencies (6)	Flips (7)	Divergences (8)	Inconsistencies (9)
				Comparison 1					
Announcement	.0758	.0917	.1335	.1129	.2055	.2821	.3490	.4098	.6007
	(.0053)	(.0058)	(.0070)	(.0107)	(.0133)	(.0150)	(.0183)	(.0177)	(.0188)
	[1,822]	[1,822]	[1,822]	[740]	[740]	[740]	[537]	[537]	[537]
Nonannouncement	.1206	.1379	.1995	.1849	.2846	.3837	.4350	.4692	.6997
	(.0142)	(.0153)	(.0176)	(.0285)	(.0336)	(.0360)	(.0384)	(.0380)	(.0356)
	[447]	[447]	[447]	[169]	[169]	[169]	[149]	[149]	[149]
				Comparison 2					
Announcement	.0982	.1209	.1707	.1298	.2525	.3281	.3776	.4555	.6505
	(.0032)	(.0035)	(.0041)	(.0056)	(.0071)	(.0077)	(.0087)	(.0082)	(.0084)
	[5,629]	[5,629]	[5,629]	[2,780]	[2,780]	[2,780]	[2,235]	[2,235]	[2,235]
Nonannouncement	.1274	.1417	.2072	.1483	.2603	.3467	.4224	.4606	.6764
	(.0032)	(.0034)	(.0040)	(.0051)	(.0062)	(.0068)	(.0074)	(.0070)	(.0069)
	[6,493]	[6,493]	[6,493]	[3,626]	[3,626]	[3,626]	[3,103]	[3,103]	[3,103]
				Comparison 3					
Announcement	.0924	.1139	.1605	.1187	.2389	.3081	.3645	.4457	.6305
	(.0028)	(.0030)	(.0035)	(.0047)	(.0060)	(.0066)	(.0076)	(.0073)	(.0075)
	[6,879]	[6,879]	[6,879]	[3,508]	[3,508]	[3,508]	[2,761]	[2,761]	[2,761]
Nonannouncement	.1211	.1350	.2003	.1399	.2557	.3392	.4095	.4476	.6656
	(.0032)	(.0033)	(.0039)	(.0049)	(.0062)	(.0068)	(.0075)	(.0071)	(.0071)
	[6,388]	[6,388]	[6,388]	[3,569]	[3,569]	[3,569]	[2,997]	[2,997]	[2,997]

NOTE.—The table displays the means (across events) of the proportions of pairwise comparisons that involve inconsistencies of the indicated type. Standard errors are in parentheses, and the numbers of observations (events) are in brackets.

Table 4: Explicit forecast revisions

10 Forecast revision evidence

10.1 Table 4: explicit forecast revisions

Table 4 studies cases in which analysts explicitly update forecasts in both windows. The paper treats these results cautiously. Explicit nonannouncement updates are likely to be selected cases in which analysts acquired private information. Therefore, the nonannouncement explicit-update sample is not a clean control sample.

Still, Table 4 shows that flips and divergences are not rare among explicit updates. For announcement-period explicit updates, the proportions of flips, divergences, and either inconsistency are economically meaningful, particularly among analyst pairs with small changes in average forecasts. The paper interprets these patterns as suggestive but not the main comparison.

10.2 Table 5: implicit forecast revisions

Table 5 is more central. Here, an analyst who does not revise is interpreted as implicitly confirming the prior forecast. Under this treatment, announcement-period inconsistencies are much more common than nonannouncement inconsistencies. The mean proportions of flips, divergences, and either inconsistency are four to five times larger than in the nonannouncement control sample in the first comparison. The longer-window comparisons strengthen the result.

TABLE 5

Average Proportions of Implicit Forecast Revisions

	All Analyst Pairs			Analyst Pairs with Small Changes in Average Forecast		
MPLE	Flips (1)	Divergences (2)	Inconsistencies (3)	Flips (4)	Divergences (5)	Inconsistencies (6)
	Comparison 1					
nnouncement	.0049	.0060	.0086	.0034	.0077	.0100
	(.0005)	(.0006)	(.0007)	(.0005)	(.0008)	(.0009)
	[7,864]	[7,864]	[7,864]	[6,641]	[6,641]	[6,641]
onannouncement	.0012	.0014	.0020	.0009	.0016	.0021
	(.0002)	(.0003)	(.0003)	(.0002)	(.0004)	(.0004)
	[8,306]	[8,306]	[8,306]	[7,545]	[7,545]	[7,545]
ifference	.0048	.0042	.0069	.0027	.0059	.0079
	(.0009)	(.0009)	(.0011)	(.0009)	(.0013)	(.0015)
	[4,147]	[4,147]	[4,147]	[3,291]	[3,291]	[3,291]

Table 5: Implicit forecast revisions, first part

	Comparison 2					
uncement	.0154	.0180	.0258	.0134	.0254	.0326
	(.0007)	(.0007)	(.0009)	(.0008)	(.0011)	(.0013)
	[11,973]	[11,973]	[11,973]	[10,032]	[10,032]	[10,032]
nnouncement	.0094	.0099	.0152	.0067	.0126	.0163
	(.0004)	(.0004)	(.0005)	(.0004)	(.0006)	(.0007)
	[14,906]	[14,906]	[14,906]	[13,866]	[13,866]	[13,866]
rence	.0076	.0103	.0139	.0093	.0172	.0224
	(.0010)	(.0010)	(.0012)	(.0012)	(.0016)	(.0018)
	[9,926]	[9,926]	[9,926]	[7,790]	[7,790]	[7,790]
	Comparison 3					
uncement	.0096	.0111	.0162	.0074	.0139	.0183
	(.0004)	(.0005)	(.0006)	(.0005)	(.0007)	(.0008)
	[15,103]	[15,103]	[15,103]	[13,608]	[13,608]	[13,608]
nnouncement	.0053	.0059	.0086	.0033	.0070	.0087
	(.0002)	(.0003)	(.0003)	(.0002)	(.0004)	(.0004)
	[16,323]	[16,323]	[16,323]	[15,614]	[15,614]	[15,614]
rence	.0068	.0078	.0115	.0069	.0115	.0159
	(.0013)	(.0012)	(.0016)	(.0013)	(.0017)	(.0021)
	[3,868]	[3,868]	[3,868]	[3,239]	[3,239]	[3,239]

E.—The table displays the means (across events) of the proportions of pairwise comparisons that involve inconsistencies of the indicated type and the mean differences in proportions. rd errors are in parentheses, and the numbers of observations (events) are in brackets.

Table 5: Implicit forecast revisions, continuation

10.3 Interpretation of Table 5

The absolute frequency of inconsistencies is small. The paper emphasizes that this is not a weakness of the theory. The model does not predict that all public announcements generate flips or divergences. It predicts that such patterns are possible when likelihoods differ, whereas identical-likelihood models rule them out. Therefore, even a small number of systematic violations is evidence against the identical-interpretation restriction.

10.4 Table 6: magnitudes of flips and divergences

Table 6 studies whether the inconsistencies are economically meaningful. Divergences average roughly 40 to 56 cents depending on the window, and flips average roughly 31 to 39 cents. Relative to the average earnings forecast in the sample, the inconsistencies are large enough to matter economically.

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TABLE 6

MEANS OF THE ABSOLUTE VALUES OF FLIPS AND DIVERGENCES

Sample	Divergences: Prior Difference (1)	Divergences: Size (2)	Divergences: Ratio (3)	Flips: Prior Difference (4)	Flips: Size (5)	Flips: Ratio (6)
	Comparison 1					
Announcement	.24	.4	.16	.41	.31	.12
	(.03)	(.03)	(.02)	(.03)	(.02)	(.01)
	[247]	[247]	[247]	[309]	[309]	[309]
Nonannouncement	.23	.4	.16	.37	.38	.15
	(.04)	(.09)	(.03)	(.05)	(.09)	(.04)
	[73]	[73]	[73]	[80]	[80]	[80]
	Comparison 2					
Announcement	.21	.43	.18	.4	.34	.14
	(.01)	(.02)	(.01)	(.01)	(.01)	(.01)
	[1,231]	[1,231]	[1,231]	[1,528]	[1,528]	[1,528]
Nonannouncement	.22	.46	.18	.38	.39	.15
	(.01)	(.02)	(.01)	(.01)	(.01)	(.01)
	[1,882]	[1,882]	[1,882]	[2,149]	[2,149]	[2,149]
	Comparison 3					
Announcement	.24	.56	.22	.46	.39	.15
	(.01)	(.01)	(.01)	(.01)	(.02)	(.004)
	[4,561]	[4,561]	[4,561]	[5,445]	[5,445]	[5,445]
Nonannouncement	.23	.54	.20	.42	.43	.17
	(.01)	(.01)	(.01)	(.01)	(.01)	(.01)
	[5,340]	[5,340]	[5,340]	[5,631]	[5,631]	[5,631]

NOTE.—Standard errors are in parentheses, and numbers of observations are in brackets.

Table 6: Magnitudes of flips and divergences

11 What exactly is identified?

11.1 Identified strongly

The paper strongly identifies a tension between observed trading volume and models in which common public information produces a common belief revision. The cleanest fact is abnormal announcement volume at or near zero returns.

11.2 Identified more cautiously

The analyst forecast evidence is powerful but more assumption-dependent. It relies on interpreting non-revisions as implicit confirmations and on comparing announcement windows to nonannouncement windows. The authors acknowledge that private information production around announcements and mechanical forecast effects can generate some inconsistencies.

11.3 Not directly identified

The paper does not prove that every inconsistency is caused by different likelihood functions. Instead, it shows that identical interpretation is too restrictive and that a different-likelihood model can rationalize both the volume-return relation and forecast-revision anomalies.

12 Economic interpretation

12.1 Disagreement versus information asymmetry

The paper separates two concepts often bundled together:

- **Information asymmetry:** agents have different information.
- **Interpretation heterogeneity:** agents have different models for mapping information into fundamentals.

The paper's central point is that public information can create trade not because agents saw different signals, but because they mapped the same signal into different posterior beliefs.

12.2 Why public news can increase trade without moving prices

Suppose one group reads an earnings announcement as good news because it believes the bad component is transitory, while another reads it as bad news because it believes the weakness is persistent. Their trades can offset in price but still generate volume. This is exactly the kind of situation the authors have in mind when describing different likelihood functions as different models of the economy or of a firm's fundamentals.

12.3 Relation to analyst disagreement

Analysts can observe the same announcement but revise forecasts differently because they disagree about persistence, margins, industry demand, cost structure, accounting quality, or managerial guidance. The paper's flips and divergences are empirical manifestations of this deeper disagreement.

13 Limitations and caveats

- **Private information around announcements:** Some abnormal volume and some analyst inconsistencies could come from private information production around earnings announcements.
- **Implicit update assumption:** Treating a non-revision as confirmation is plausible but not perfect. Analysts may not update because updating is costly.
- **Mechanical flips:** Because analysts forecast annual earnings and quarterly news mechanically changes remaining-year arithmetic, some flips can occur without genuine interpretation differences.
- **Model simplification:** The theoretical model takes heterogeneous likelihoods as primitive. It does not fully derive why agents have different likelihood functions from deeper learning histories.

14 Conclusion

Kandel and Pearson provide an early and influential empirical and theoretical argument that public information does not necessarily mean common interpretation. Around earnings announcements, trading volume is abnormally high even when prices do not move. Analysts' forecasts also show patterns that identical-likelihood Bayesian updating rules out. The paper's different-likelihood model gives a simple explanation: public signals can create disagreement, not merely resolve it.

The paper's broader contribution is to make interpretation itself an object of financial economics. In markets, agents do not only differ in what they know; they may differ in how they process what everyone knows.

15 Short takeaway

> The paper shows that public signals can generate speculative trade because agents interpret the same news differently; prices may barely move because beliefs offset, while volume rises because disagreement increases.